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# New paleomagnetic results from the ca. 1.68–1.63 Ga mafic dyke swarms in Western Shandong Province, Eastern China: Implications for the reconstruction of the Columbia supercontinent

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ABSTRACT

The existence of Paleo-Mesoproterozoic supercontinent *Columbia* (aka Nuna) was established decades ago, but the position of North China Craton within *Columbia* is still highly debated due to the paucity of available high-quality paleomagnetic and reliable geological constraints. Precise geochronological dating of extensive Mesoproterozoic mafic dyke swarms in Western Shandong Province (also named the Luxi area), China reveals two phases of dyke emplacement at ~1.68 Ga and ~1.63 Ga. In this paper, we report new paleomagnetic and rock magnetic results obtained from approximate 160 samples (16 sites) collected from these two phases of mafic dyke swarms in the Luxi area with the aim of pinpointing the location of North China Craton within *Columbia* supercontinent in this time interval. Rock magnetic experiments confirm that either magnetite or titanomagnetite is the main magnetic carrier in these dykes. Stepwise thermal demagnetization revealed two paleomagnetic poles. For the ~1.63 Ga dykes, normal and reversed high-temperature remanent magnetization directions yield a mean direction (*D/I*) of 86.1°/53.5° ($\kappa$ = 43.3, $\alpha_{95}$ = 7.9°, $N$ = 9). These directions pass a reversal test and are interpreted as primary remanences. The corresponding paleomagnetic pole is calculated at 20.8°N, 182.5°E ($\kappa$ = 28.3, $A_{95}$ = 8.3°, $N$ = 9). This pole passes the examination of secular variation of geomagnetic field (Deenen et al., 2011, 2014). It fulfills a Van der Voo (1990) value Q = 6 and is therefore suggested to be a 'key' pole demarcating the Precambrian North China Craton. For the ~1.68 Ga dykes, only the normal directions are isolated with a mean direction (*D/I*) of 89.1°/47.1° ($\kappa$ = 35.0, $\alpha_{95}$ = 13.1°, $N$ = 5) with a corresponding paleomagnetic pole of 17.8°N, 184.9°E ($\kappa$ = 29.6, $A_{95}$ = 14.3°, $N$ = 5). This pole passes the examination of secular variation of geomagnetic field. Finally we select the ~1.63 Ga high-quantity paleomagnetic pole in order to depict a more detailed apparent pole wander path (APWP) to compare with the other major Precambrian cratons. Combined with other geological evidence, our reconstruction scenario supports the spatio-temporal connection between the Baltica, North Australian Craton and North China Craton.

## 1. Introduction

The Paleoproterozoic supercontinent *Columbia* (aka Nuna), was first proposed because of numerous Paleoproterozoic orogens on most Precambrian continents (Zhao et al., 2002; Roger and Santosh, 2002). After its amalgamation, *the Columbia* supercontinent became relatively stable during the early Mesoproterozoic (Cawood and Hawkesworth, 2014). Geological activity in this period appears to have been limited to intra-plate magmatism consisting of widespread emplacement of mafic dyke swarms and sills across many continents. The abundance of these geological units has been the focus of much researches demonstrating the locations of ancient cratons, establishing the timing of their activity, and allowing theories of supercontinent extent to be developed and tested (Halls et al., 2000; Klein et al., 2016; Salminen et al., 2014, 2017, 2018; Buchan et al., 2000; Pisarevsky et al, 2014; Irving et al., 1972, 2004; Harlan et al., 2008; Buchan and Halls, 1990; Veselovskiy et al., 2006; Ernst and Buchan, 2000; Shankar et al., 2018; Zhang et al., 2009, 2012b; Chen et al., 2013; Meert et al., 2011).

A variety of models has been proposed regarding the evolution of *the Columbia* (Zhao et al., 2002; Roger and Santosh, 2002; Johansson,

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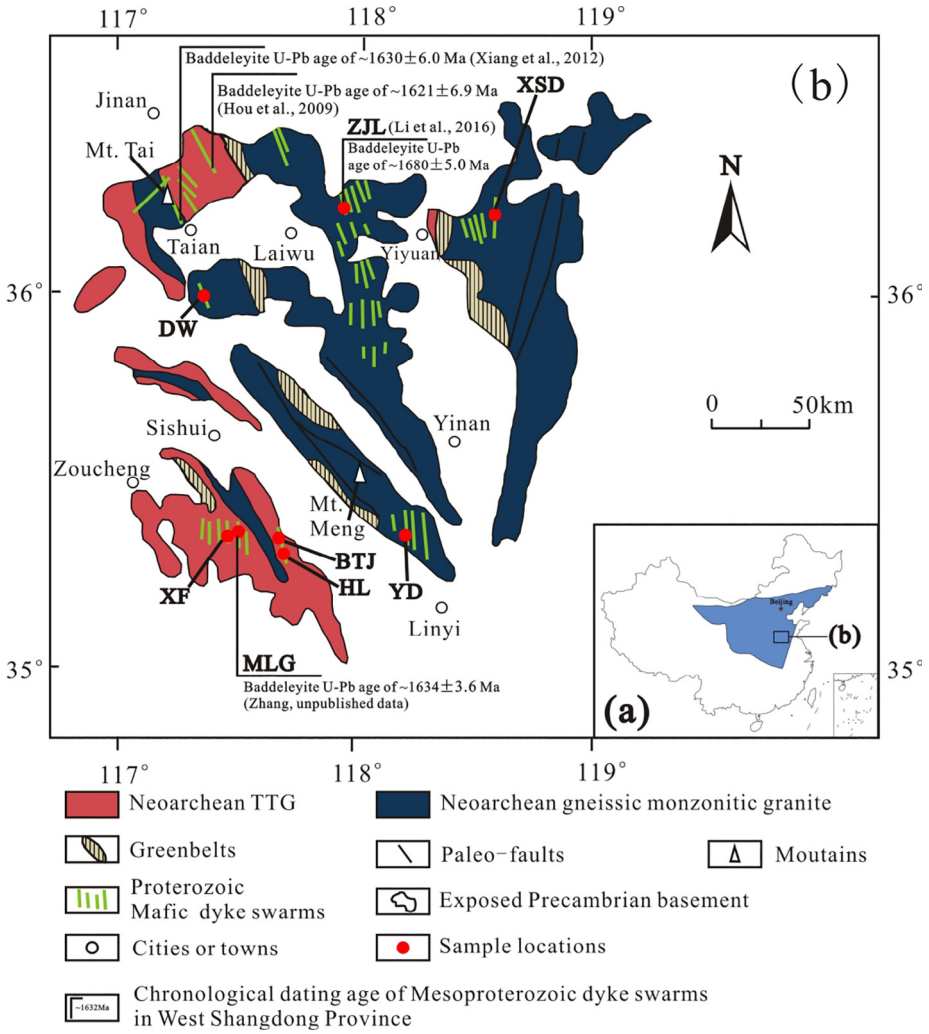



**Fig. 1.** (a) The location of the North China Craton and the Luxi study area, Western Shandong Province, China. (b) Geological sketch map of ca. 1.68–1.63 Ga mafic dyke swarms (modified from Hou et al., 2009). Sample sites are shown by closed red circles. Previously published dyke ages are also displayed with their references. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2009; Evans and Mitchell, 2011; Zhang et al., 2012a; Pisarevsky et al. 2014) but the location of North China Craton (NCC) has yet to be established. For the adjacent continents to the north of NCC, India (Zhao et al., 2002; Zhang et al., 2012a; Xu et al., 2014), Baltica (Wilde et al., 2002; Pei et al., 2005), Laurentia (Halls et al., 2000; Wu et al., 2005), Siberia (Halls et al., 2000; Evans and Mitchell, 2011; Chen et al., 2013; Pisarevsky et al. 2014) and North Australia Craton (Zhang et al., 2012a; Pisarevsky et al., 2014; Xu et al., 2014; Zhang et al., 2017, 2018) were all proposed as possible candidates. There are only 6 reliable Mesoproterozoic paleomagnetic poles constraining the NCC (Table 2) rendering its location within *the Columbia* ambiguous. Pisarevsky et al. (2014) have summarized a quantity of paleomagnetic data in time-space distribution for elaborating reconstruction, in which massive paucity of paleomagnetic data was presented for NCC, especially during 1.78–1.50 Ga. A number of more recent geochronological studies revealed Mesoproterozoic mafic dyke swarms in Western Shandong Province intruded at ca. 1.68–1.63 Ga, which would potentially and crucially be ideal objects for paleomagnetic studies to fill in this gap (Hou et al., 2009; Xiang et al., 2012; Li et al., 2015; Zhang Shuan-hong, unpublished data). Here we examine such precisely dated mafic dyke swarms (ca. 1.68–1.63 Ga) in Western Shandong Province in terms of paleomagnetism and rock magnetism. The acquisiton of such data is essential for the Paleo-Mesoproterozoic supercontinent *the Columbia*. Our aim is to determine the position of the North China Craton within *the Columbia* supercontinent and to find its spatio-temporal relations with other adjacent continents.

## 2. Geological setting

The NCC can be roughly divided into three main tectonic units: the Eastern Block, the Trans-North China Orogeny (TNCO) and the Western Block (Zhao et al., 2005). The Eastern and Western blocks collided and combined along a linear TNCO at ~1.85 Ga, which sets a maximum age of the formation of united North China Craton. As one of the oldest cratons, the Precambrian metamorphic crystallized basement rcok is mainly exposed on the TNCO, the northern margin and the Eastern Block of the NCC. In the Eastern Block of NCC, the exposed metamorphic basement rocks are composed of Tonalite-Trondhjemite-Granodiorite (TTG) gneiss, gneissic monzonitic granite and amphibolites, which are surrounded by greenstone belts. Specifically, Paleo-Mesoproterozoic sedimentary rocks are well developed in the Yanliao area in the Eastern Block of NCC, resting unconformably on the underlying Archean metamorphic basement rock. These Paleo-Mesoproterozoic strata are generally divided into Changcheng, Jixian and Daijian Systems from the base to top. They consist of conglomerates, sandstones, shales, clastic and carbonate rocks which develop in extensional settings or neritic environments. These Paleo-Mesoproterozoic strata were subjected to numerous magma intrusions during this

period.

Reliable dating of both igneous and sedimentary rocks is essential to temporally contextualize the geological events of NCC. The Changcheng System divides into the Changzhougou, Chuanlinggou, Tuanshanzi and Dahongyu Formations from the base to top. Although the age of the lower part of the Changcheng System is questionable, zircon LA-MC-ICP-MS U-Pb dating of a granoporphyric vein from Miyun, Beijing revealed that the age of the lower limit must be less than 1.67 Ga (Li et al., 2011). Zhang et al. (2013) also applied LA-ICP-MS U-Pb dating to a diorite-porphyrite vein intruding the Chuanlinggou Formation, which revealed that the depositional age of the Chuanlinggou Formation is less than 1.63 Ga. Futher LA-ICP-MS U-Pb obtained from the upper potassium-rich volcanic rock of the Tuanshanzi Formation indicated that the sedimentary age of the upper part of the Tuanshanzi Formation was ~1.64 Ga. A extensive (up to 600 $km^2$) thick (718 m) potassic volcanic rock layer in the upper part of the Dahongyu Formation (Hu et al., 2007) has provided futher chances for dating. For example, Gao et al. (2008) employed zircon SHRIMP U-Pb dating to this volcanic layer which yielded an emplacement age of ~1.63 Ga which overlaps (within error bars) with previous studies (Lu and Li, 1991; Lu et al., 2008).

In addition to the ~1.63 Ga volcanic rocks, several stratigraphically coeval mafic dyke swarms have recently been also identified, a number of which occur in Western Shangdong Province (WS). This province, also referred to as the Luxi area, is separated from the Eastern Shandong Province (ES) by the Tanlu Fault. WS is famous for its large distribution of exposed Archean basement rocks and Neoarchean greenstone belts (Fig. 1; Xu et al., 1992; Wan et al., 2011). The Archean basement rocks in WS are primarily composed of TTG gneiss and gneissic monzonitic granite, which are overlapped in places by horizontally-bedded Cambrian limestones (Fig. 2a), thus implying that Phanerozoic tectonism had little impact in this area.

Large scale of mafic dyke swarms are widely distributed near Tai'an, Laiwu, Yiyuan, Zoucheng and Linyi cities in WS (Fig. 1). Most of the outcrops exhibit minimal weathering, deformation or metamorphism. These mafic dyke swarms intruded into the Archean basement rocks and were then overlapped by near horizontal Cambrian limestones in many places (sample site XSD in the eastern part of Yiyuan town, Fig. 2a, c and d). Accurate age constraints on the emplacement age of these mafic dyke swarms in WS have been presented previously. For example, Hou et al. (2009) reported an emplacement age of ~1620.8 ± 6.9 Ma using baddeleyite TIMS U-Pb dating from a dyke in western Laiwu. Xiang et al. (2012) applied both ID-TIMS and SIMS U-Pb methods to a mafic dyke near Tai'an which yielded ages of ~1621.1 ± 8.8 Ma and ~1632.4 ± 4.2 Ma, respectively. Li et al. (2015) reported an somewhat older emplacement age of ~1680 ± 5.0 Ma from the eastern Laiwu dyke (ZJL dyke) using baddeleyite SIMS U-Pb dating. Zhang Shuan-Hong (unpublished data) obtained an emplacement age of ~1634 ± 3.6 Ma also using the baddeleyite SIMS U-Pb method from a mafic dyke in eastern Zoucheng, consistent with others' works. In summary, these studies attest to the intrusion of mafic dyke swarms from ca. ~1.68–1.62 Ga.

## 3. Field occurrence and sampling

Early Mesoproterozoic mafic dyke swarms in WS were investigated. Drilled samples for paleomagnetic measurements were taken from unweathered rocks at all sites. The ~1.68 Ga mafic dyke, labelled ZJL, which has intruded into Archean crystallized basement rocks was sampled. The ZJL dyke is 10 m wide, trenching N-S (0°E) and is subvertical, dipping east (75°) (Li et al., 2015). The petrology of this dyke is typically dolerite with a mineral assemblage comprised primarily of clinopyroxene and plagioclase but has undergone varied saussuritization (Li et al., 2015).

Moreover, another 10 m wide dyke oriented NNE (15°) (herein referred to as XSD) overlapped by horizontally-bedded Cambrian limestones is clearly exposed on both sides of a road in the eastern Yiyuan town (Fig. 2a, c and d). Roks in this outcrop exhibit more vertical joints compared to dyke ZJL, but only exhibit spheroidal weathering in places. The XSD dyke intruded into Neoarchean monzonitic granite, which shows sharp contact with the country rock yet the boundary is vertical (Fig. 2c).

Four sites from west to east (termed as XF, MLG, BTJ and HL) were also selected to be investigated in eastern Zoucheng city. Dyke XF is 15 m and has a trending of NNW (340°) with an eastward dip of 75°. The MLG dyke is 10 m wide, trends NNW (340°), dips (78°) to the west and is intruded into Archean basement rocks resulting in a sub-vertical stratigraphic contact (Fig. 2b). Zhang Shuan-hong (unpublishe data) obtained a preliminary age estimate of ca. ~1.63 Ga from this dyke.

A single large unweathered dyke (at least 5 km long and 17 m wide) is presented at sites BTJ and HL. This dyke trends in a NNW (340°) direction. The boundaries between the dyke and the Archean monzonitic granite are consistently sharp and vertical.

Another dyke referred to as YD in the northern part of Linyi City was also sampled. Similar to the dykes near Zoucheng City, the YD dyke trends in a N-S direction, exhibit a vertical boundary with the country rock and is 12 m wide. The YD dyke experienced only very thin weathering depth.

A dyke (referred to as DW) in the southern part of Tai'an City was also sampled; however, it exhibits no boundary with the country rocks rendering estimation of the trend, dip and width of the dyke problematic. Using geological map, we tentatively suggest DW dyke trends N-S. Previous studied dykes in this area were dated to ~1.63 Ga (Xiang et al., 2012).

Approximate 160 paleomagnetic samples were thus collected by portable drilling form the locations shown in Fig. 1. Sampling strategy ensured representation of all the dykes outlined in the above text. Samples were taken aligned to the trending of each dyke (covering an area of 10000 $km^2$) to maximize the reliability of the resultant paleomagnetic data.

All samples were oriented using sun compass to correct for large angular deviation caused by the intense magnetization exhibited by diabase which has strong impacts on magnetic compass. The differences in measured orientation between these two devices may range from 60° to 110°.

## 4. Methods

### 4.1. Rock magnetic experiments

The nature of magnetic minerals can be detected through rock magnetic experiments to evaluate whether existing magnetic minerals in a given rock can carry primary information of paleomagnetic field. Thus, based on lithology and thermal demagnetization results, a series of rock magnetic experiments were undertaken to determine the magnetic minerals composition of Mesoproterozoic mafic dyke swarms in the Luxi area. These included thermomagnetic experiments, thermal demagnetization of three-component IRM experiments, magnetic hysteresis loops and first order reversal curves (FORCs). Several specimens from the ~1.68 Ga and ~1.63 Ga mafic dykes were selected to complete these experiments, in which sample ZJL4-9 (Li et al., 2015) and sample MLG6-5 (Zhang Shuan-hong, unpyblished data) are precisely dated.

Thermomagnetic (κ-t) experiments, magnetic hysteresis loops and first order reversal curves (FORCs) were all carried out at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Natural Resources (Beijing). For Thermomagnetic experiments, selected samples were ground into flour, placed into a cuvette and then measured by a CS-4 instrument. This instrument measures variations in the magnetic susceptibility exhibited by magnetic minerals when subjected to heating up to 700 °C and then cooling down back to room temperature in an oxygen environment. Magnetic hysteresis loops and FORCs can reveal both the grain size and domain state of magentic

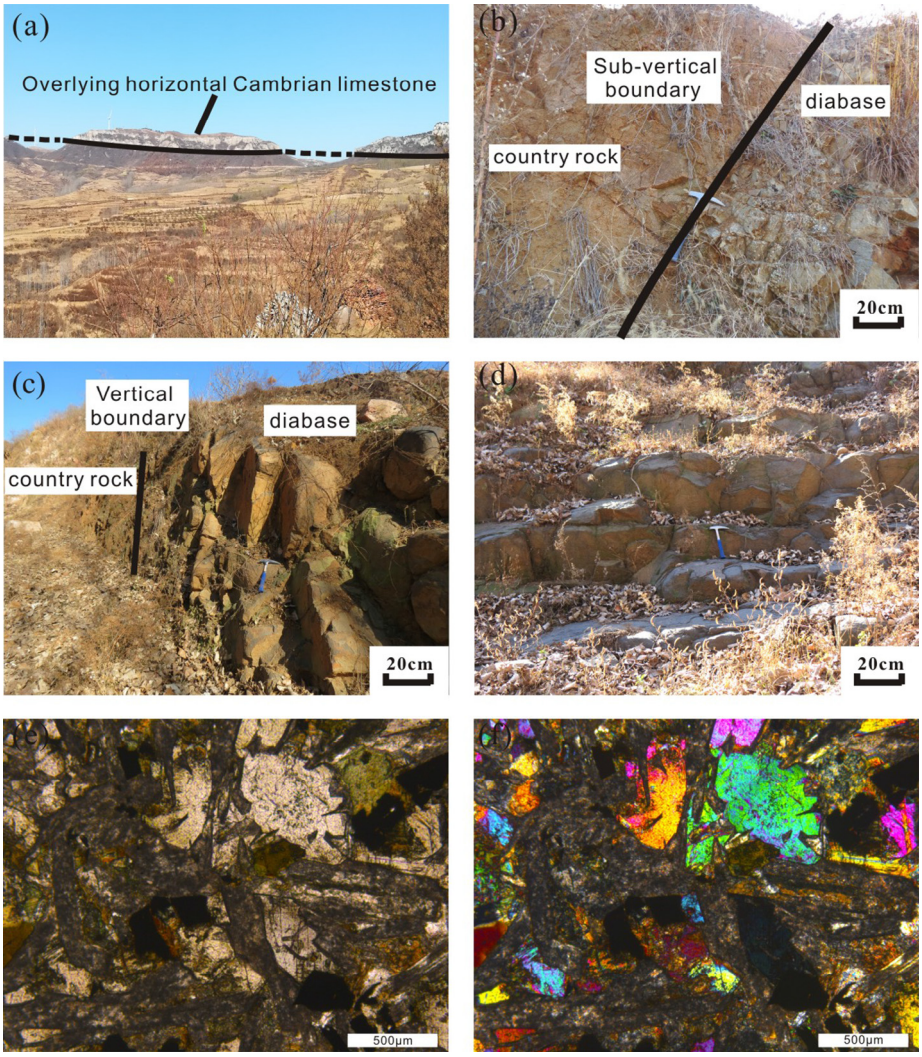



**Fig. 2.** Field Photographs showing (a) Conformable contact between Archean basement rock and overlying horizontal Cambrian limestone. (b) Sub-vertical boundary between the country rock and diabase dyke at site MLG. (c-d) Vertical boundary between the country rock and dyke at site XSD and its corresponding outcrop. (e-f) Photomicrograph of selected samples from site BTJ.

minerals and were obtained using a Lakeshore 8600 Vibrating Sample Magnetometer (VSM). For the magnetic hysteresis loop experiments, the powdered samples were subjected to a cycled field of ± 700 mT. A slope correction was applied to remove the impact of paramagnetic contributions. For FORCs (Roberts et al., 2000), an average time of 0.1 s was accepted during the measurements, after which FORCinel software (Harrison and Feinberg, 2008) was used to process the data.

Thermal demagnetization of three-component isothermal remanent magnetization (IRM) experiments were conducted in Nanjing University (Nanjing). This approach is used to reveal the unblocking temperature of magnetic minerals (Lowrie, 1990). Firstly, DC fields of 2.5 T, 0.4 T and 0.12 T were successively applied on Z axis, Y axis and X axis by an ASC pulse magnetizer. Secondly, the selected standard paleomagnetic specimens were subjected to stepwise thermal demagnetization and were measured by an AGICO JR-6A spinning magnetometer. The temperature setting was the same as the thermal demagnetization of NRM which will be outlined below.

### 4.2. Thermal demagnetization experiments

Usually, stepwise thermal demagnetization of natural remanent magnetization (NRM) is necessary to acquire a geomagnetic direction in a rock. This experiment was conducted in a magnetically shielded house to avoid the impact from the modern geomagnetic field. All measurements were carried out in Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Natural Resources (Beijing). All fresh samples were cut into standard paleomagnetic specimens and demagnetized in an ASC TD-48 thermal demagnetizer. After each heating step, the remanence was measured using 2G-755 cryogenic magnetometer or AGICO JR-6A spinning magnetometer. Experiments on pilot specimens yielded suitable results when heated from 100 °C to 600 °C with 16 heating steps. As such, this measurement protocol was applied to all remaining samples. Temperature intervals were set at 30–50 °C for temperatures lower than 480 °C and reduce to 20 °C at temperatures higher than 480 °C. As for data processing, principal component analysis (PCA) (Kirschvink, 1980) in conjunction with the best-fit great circles method (McFadden and McElhinny, 1988) were employed to calculate paleomagnetic results. Fisher statistics (Fisher, 1953) and/or the intersection of great circles (McFadden and McElhinny, 1988) were applied to address site-mean directions.

## 5. Results

### 5.1. Rock magnetic results

Representative specimens yield satisfactory rock magnetic results

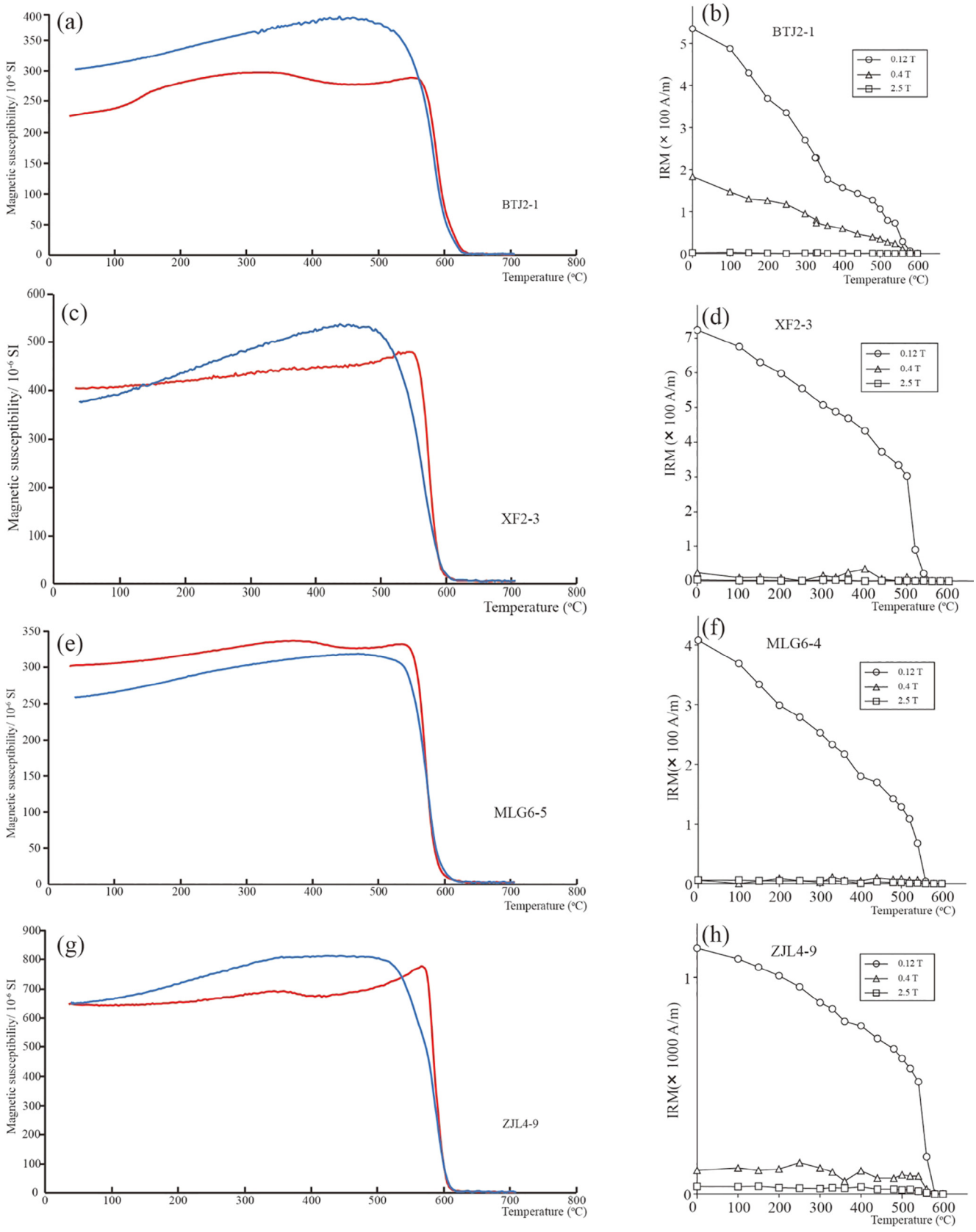



**Fig. 3.** Resutls of rock magnetic experiments from representative samples. (a, c, e and g) Thermomagnetic curves (susceptibility versus temperature) from representative samples. Red curves represent the heating curves and blue curves the cooling curves. All thermomagnetic curves are measured in air. (b, d, f and h) Results of thermal demagnetization of three-component isothermal remanent magnetization (IRM) experiments from 4 samples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(Fig. 3). In thermomagnetic experiments, the heating and cooling curves of these specimens are nearly reversible. The heating curve of these specimens present an insignificant rise in magnetic susceptibility at 350 °C; however, a rapid increase after heating to 550 °C. At ~580 °C, the value of magnetic susceptibility of the heating curve attained its maximum level and then dramatically dropped down to zero at ~600 °C, forming a Hopkinson peak, best expressed by the sample ZJL 4–9 (Fig. 3g). This pronounced peak and the unblocking temperature imply the existence of magnetite. The thermal demagnetization of three-component IRM reveals that the soft magnetic component dominates over the medium and hard magnetic components. The intensity of soft magnetic component gradually slows down until 500 °C with an inflexion at 500 °C. After 500 °C, the intensity of soft magnetic component rapidly drops and comes to zero at ~580 °C (Fig. 3b, 3f and 3 h), indicative of the unblocking temperature at ~580 °C; however, the unblocking temperature of soft magnetic component of specimen XF2-3 is at ~560 °C, implying the existence of titanomagnetite (Fig. 3d). Overall, results from thermal demagnetization of three-component IRM confirm the existence of magnetite and imply that there may be titanomagnetite as well. An additional inflexion at ~350 °C of the medium component of ZJL4-9 also attests to the presence of Fe-sulfides (Fig. 3h). Results from the κ-t curve experiment also indicate a small increase in magnetic susceptibility at 350 °C (Fig. 3g) and may corroborate this interpretation. This may be explained by the fact that Fe-sulfides, like pyrrhotite, will be transformed into magnetite when heated in air (Dunlop and Özdemir, 1997).

For the magnetic hysteresis experiments, a representative sample MLG6-5 exhibits a narrow-waisted hysteresis loop with a saturation field of approximate 400 mT and a coercivity of 12.5 mT (Fig. 6a), also typical of (titano)magnetite (David, 1983). Besides, magnetic hysteresis parameters are used for Day diagram (Day et al., 1977; Dunlop, 2002), which is provided in Fig. 6d. Five representative samples from both the ~1.68 Ga and ~1.63 Ga mafic dykes all plot in pseudo-single domain (PSD) sector while FORC diagrams from two representative samples MLG6-5 and ZJL4-9 reveal the grain size of the (titano)magnetite to be single domains (SD) (Fig. 6b and 6c).

To conclude, Hopkinson peak in conjunction with the unblocking temperature at ~580 °C or ~560 °C indicate magnetite or titanomagnetite constitutes the primary magnetic mineral within dykes and there may be another potential magnetic mineral like pyrrhotite. Magnetic hysteresis experiments and FORCs unambiguously demonstrate this assemblage is SD or PSD.

### 5.2. Paleomagnetic results

Results of thermal demagnetization are plotted in orthogonal projections (Zijderveld, 1967) as well as equal-area projections (Fig. 4). For most specimens, the intensity of the remanent magnetization gradually decreases below 500 °C and drops more rapidly above 500 °C, indicating the presence of remanence-carrying magnetite. This is consistent with rock magnetic results. The intensity of remanence of sample XSD1-4 fluctuates below ~300 °C; however, it also drops sharply when heated above 500 °C. The remanent magnetization of the specimens displayed in Fig. 4 reaches to zero within the temperature ranging from 500 °C to 600 °C. Thus, the Curie temperature attests to the presence of magnetite.

For those specimens whose intensity are commonly larger than 1 A/m, for example, specimen BTJ2-1 (Fig. 4a), the vector of remanence is expressed by a straight line towards the origin in the orthogonal projection above ~100 °C. Only in the first step of demagnetization the remanence vector is inconsistent with the other steps (Fig. 4a). To avoid the influence of viscous remanent magnetization, the components are isolated at temperatures above 480 °C yielding a direction (*D/I*) of 281.8°/−41.2° ($\alpha_{95}$ = 0.6°). Other specimens from site BTJ2 can also isolate such a stable direction. The mean declination of site BTJ2 is 275.4° while the mean inclination of site BTJ2 is −52.1° ($\kappa$ = 24.9, $\alpha_{95}$ = 11.3°) (Table 1). With regard to ZJL4-9 (Fig. 4e), its demagnetization features are similar to that of BTJ2-1. Notably a high-temperature component is evident above 540 °C, the direction of which is likely antipodal to that of BTJ2-1; however, only two steps (step 560 °C and step 580 °C) are available to calculate the direction. Thus, the best-fit great circles method is applied in such cases. Another specimen ZJL4-2 from the same site exhibits a direction (*D/I*) towards the origin of 94.7°/43.3° ($\alpha_{95}$ = 7.1°) from step 500 °C to step 600 °C as calculated by PCA. For site ZJL4, the mean direction (*D/I*) is 94.0°/44.7° ($\kappa$ = 348.8, $\alpha_{95}$ = 3.9°) (Table 1).

For specimen MLG6-5 (Fig. 4b), a stable high-temperature component is isolated above 300 °C (*D/I* is 10.9°/46.3°, $\alpha_{95}$ = 2.4°) whereas the mean declination and inclination of site MLG6 are 82.5°/47.6° with a relatively higher value of $\alpha_{95}$ which is 16.5° ($\kappa$ = 22.5) (Table 1). From step NRM to step 540 °C, a 'pin' shape is generated whereby the plotted point of step 540 °C returns to that of the first step of demagnetization. Sample XF2-3 does not present a direction towards the origin in the orthogonal projection (Fig. 4f). Meanwhile, a significant curve occurs from step 100 °C to step 500 °C yet shows an trend analogous to that observed for sample MLG6-5 in that it returns to the second quadrant (Fig. 4b and 4f). In conjunction with other specimens, a mean direction (*D/I*) is 85.4°/57.3° ($\kappa$ = 74.7, $\alpha_{95}$ = 8.1°) for site XF2 (Table 1). In addition, both of sites MLG6 and XF2 exhibit normal polarity in the equal-area projection.

Specimen XSD1-4 exhibits both normal and reversed polarity in the equal-area projection (Fig. 4c). The directions obtained at low temperatures plot along a great circle from the NRM to 330 °C. A middle-temperature component is evident from step 330 °C to step 540 °C with a high-temperature component in evidence above step 580 °C; however, it's not possible to confidently define these components and calculate a direction on the basis of only two steps. Thus, the best-fit great circles method is also applied to this specimen. A mean direction (*D/I*) of 77.3°/39.3° ($\kappa$ = 328.6, $\alpha_{95}$ = 4.4°) (Table 1) is obtained for site XSD1. This site also presents a similar trend above step 560 °C with ZJL4-9 in the equal-area projection (Fig. 4c and e).

The thermal demagnetization characters of specimen YD1-6 are similar to that of specimen ZJL4-9 (Fig. 4g). In the Zij diagram, the vector of remanence is presented by a near-straight line but it plots in a part of a great circle in the steroplot. Therefore, the best-fit great circles method is applied to this specimen. On this basis, the mean direction (*D/I*) of 71.1°/59.5° ($\kappa$ = 84.7, $\alpha_{95}$ = 6.1°) (Table 1) is obtained for site YD1.

Although directions different from the modern geomagnetic field are isolated for most of the measured specimens, specimen DW5-11 represents an exception whereby the direction of the remanence vector has been overprinted by modern geomagnetic field. For this specimen, two components can be isolated (Fig. 4d). One is a low-temperature great circle ranging from step NRM to step 400 °C and the another one is a high-temperature component from step 480 °C to step 520 °C with a direction (*D/I*) of 340.5°/54.8° ($\alpha_{95}$ = 2.2°) – close to that of the modern geomagnetic field in the study area. Therefore, it is reasonable to conclude that specimen DW5-11 carries a viscous remanent magnetization.

To summarize, the site-mean directions of 16 sites are listed in Table 1, except for site DW5 which was totally remagnetized. It is possible to isolate stable directions within acceptable errors for 16 sites; however, for sites MLG6 and XSD4, the $\alpha_{95}$ values are higher than 15° and should be treated with caution.

## 6. Discussion

Hou et al. (2009) have carried out on both geochronological and paleomagnetic investigations on mafic dyke swarms in the Luxi area; however, due to the limited number of samples in their study, the reliability of the calculated paleomagnetic pole for NCC at ~1.62 Ga is questionable; however, the results from our study demonstrate

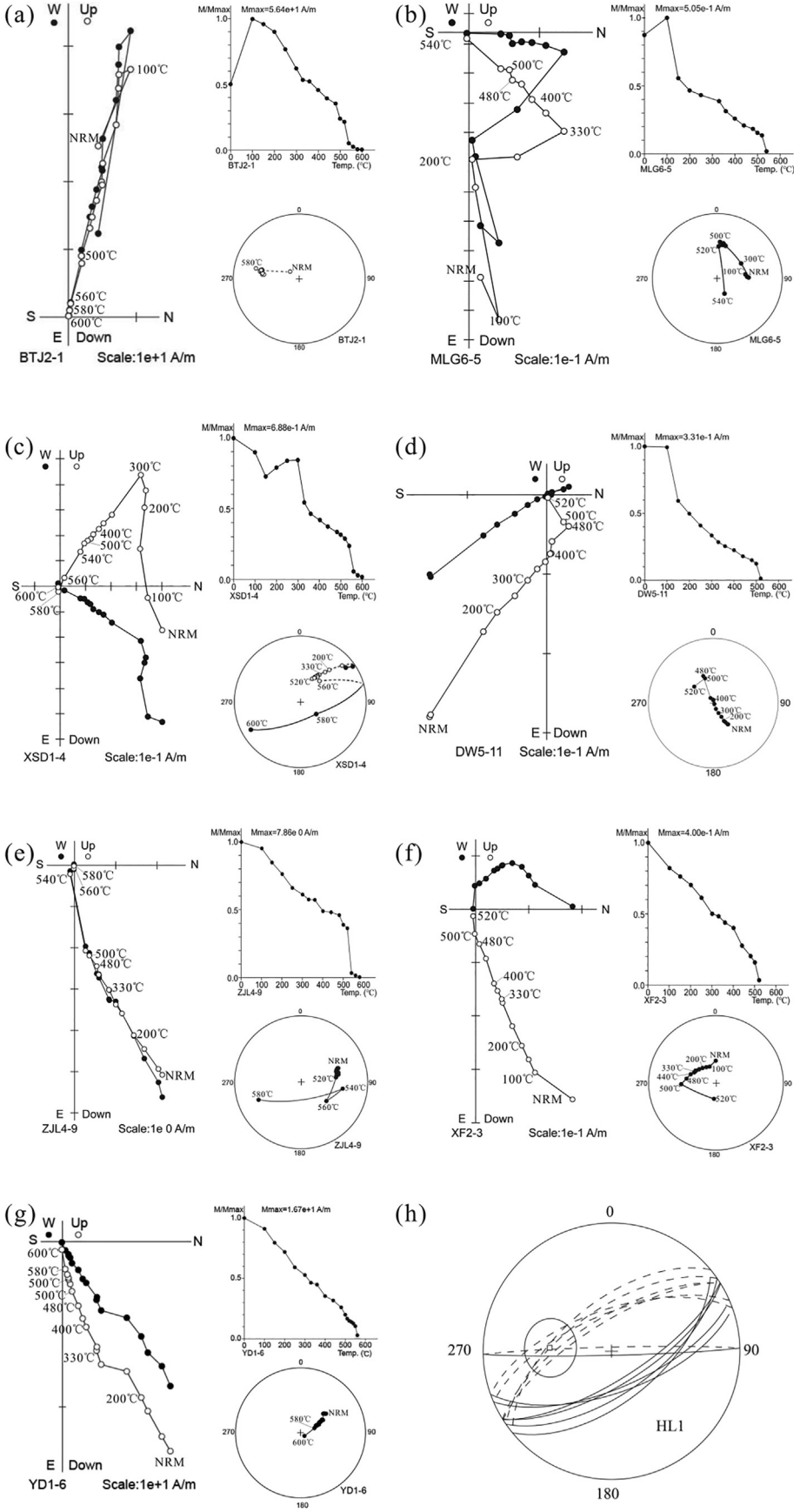



**Fig. 4.** (a-g) Representative samples showing results of stepwise thermal demagnetization. All the specimens are plotted in-situ. In the Zijderveld diagram (orthogonal vector plots), open (closed) circles represent projections on the vertical (horizontal) plane. In the steroplots, open (closed) circles represent reversed (normal) inclination. Temperature procedures are shown besides the symbols. (h) Example of site HL1 to present the intersection of great circles. Open squares represent site-mean direction of site HL1 by the intersection of the great circles.

**Table 1**
Paleomagnetic results from ~1.68 to 1.63 Ga mafic dyke swarms in the Luxi area.

| Site | Trend (°) | Dip (°) | Width (m) | n | $D$ (°) | $I$ (°) | $\kappa$ | $\alpha_{95}$ (°) | Plat. (°N) | Plon. (°E) | Dp/Dm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *~1.63 Ga mafic dykes (in Western Luxi area)* | | | | | | | | | | | |
| BTJ2 | 340 | 90 | 17 | 8 | 275.4 | −52.1 | 24.9 | 11.3 | −0.6 | 173.1 | 9.5/14.9 |
| BTJ3 | 340 | 90 | 17 | 5 | 268.8 | −39.7 | 44.6 | 11.6 | 13.7 | 189.3 | 8.4/13.9 |
| HL1 | 340 | 90 | 17 | 7 | 274.2 | −51 | 144.1 | 6.1 | 11.0 | 184.5 | 12.1/19.5 |
| HL2 | 340 | 90 | 17 | 6 | 252.6 | −47.4 | 260.4 | 5.4 | 23.2 | 192.5 | 8.5/14.1 |
| $MLG3_a$ | 340 | 78 W | 10 | 6 | 112.4 | 56.5 | 273.4 | 5.3 | 8.1 | 167.9 | 12.0/16.8 |
| $MLG5_a$ | 340 | 78 W | 10 | 5 | 77.8 | 40 | 108.8 | 10.8 | 30.4 | 189.7 | 6.3/9.4 |
| $MLG6_a$ | 340 | 78 W | 10 | 5 | 82.5 | 47.6 | 22.5 | 16.5 | | | |
| $MLG7_a$ | 340 | 78 W | 10 | 7 | 56.7 | 51 | 344.8 | 3.7 | 43.2 | 194.7 | 3.4/5.0 |
| XF1 | 340 | 75E | null | 8 | 99.7 | 60 | 91 | 6.8 | 16.0 | 168.1 | 7.8/10.3 |
| XF2 | 340 | 75E | null | 7 | 85.4 | 57.3 | 74.7 | 8.1 | 24.0 | 176.6 | 8.6/11.8 |
| Mean | | | | $N = 9$ | 86.1 | 53.5 | 43.3 | 7.9 | 20.8 | 182.5 | $A_{95}$ (°) = 8.3 |
| *~1.68 Ga mafic dykes (in Eastern Luxi area)* | | | | | | | | | | | |
| XSD1 | 15 | 90 | 10 | 6 | 77.3 | 39.3 | 328.6 | 4.4 | 22.8 | 196.9 | 3.1/5.3 |
| XSD2 | 15 | 90 | 10 | 12 | 89.1 | 52.5 | 33.4 | 7.8 | 19.5 | 181.2 | 7.4/10.7 |
| XSD4 | 15 | 90 | 10 | 3 | 85.4 | 59.2 | 53.2 | 17.1 | | | |
| $ZJL3_b$ | 0 | 78E | 10 | 7 | 107.2 | 35.8 | 82 | 8.2 | −1.3 | 181.9 | 5.5/9.5 |
| $ZJL4_b$ | 0 | 78E | 10 | 6 | 94 | 44.7 | 348.8 | 3.9 | 12.2 | 184.1 | 3.1/4.9 |
| YD1 | 0 | 90 | 12 | 8 | 71.1 | 59.5 | 84.7 | 6.1 | 35.2 | 180.1 | 6.9/9.2 |
| Mean | | | | $N = 5$ | 89.1 | 47.1 | 35.0 | 13.1 | 17.8 | 184.9 | $A_{95}$ (°) = 14.3 |

Notes: n-number of demagnetized samples for calculation of directions of each site; $D$-site-mean declination in stratigraphic coordinates; $I$-site-mean inclination in stratigraphic coordinates; $\kappa$-precision parameter of Fisher (1953); $\alpha_{95}$-radius of the 95% confidence circle on the mean direction; Plat.-paleo-latitude of calculated virtual geomagnetic poles in stratigraphic coordinates; Plon.-paleo-longitude of calculated virtual geomagnetic poles in stratigraphic coordinates; $_a$-emplacement age of ~1.63 Ga mafic dyke MLG by baddeleyite SIMS U-Pb dating from Zhang Shuan-hong, unpublished data; $_b$-emplacement age of ~1.68 Ga mafic dyke ZJL by baddeleyite SIMS U-Pb dating from Li et al., 2015.

convincing paleomagnetic poles of NCC at ~1.68 Ga and ~1.63 Ga in Luxi area.

According to previous studies, dykes in both the Western Luxi area and Eastern Luxi area exhibit differently either in the emplacement ages (Hou et al., 2009; Xiang et al., 2012; Li et al., 2015; Zhang Shuan-Hong, unpublished data) or geochemistry (Hou et al., 2005; Xiang et al., 2012; Li et al., 2015; Zhang Shuan-Hong, unpublished data). This implies that there were two stages of magmatic intrusion in the Luxi area. To address this, we will discuss our new paleomagnetic data in two groups: (i) ~1.63 Ga dykes in Western Luxi area (including dykes DW, XF, MLG, BTJ (HL) that outcrop near Tai'an and Zoucheng City); (ii) and the other from ~1.68 Ga dykes in the Eastern Luxi area (including dykes ZJL, XSD and YD that outcrop in Eastern Laiwu and near Linyi City).

### 6.1. Paleomagnetic pole of North China Craton at ~1.63 Ga

The mafic dyke swarms dated at ~1.63 Ga in the Western Luxi area were apparently not subjected to Phanerozoic tectonism on the basis of the morphology of the dykes (more than 10 m wide), sharply vertical to sub-vertical boundaries and horizontally overlapping Cambrian strata (Fig. 2a). In addition, the coeval Satakunta dyke swarms (~1.64 Ga) in Finland (Salminen et al., 2014, 2017) which exhibit similar properties to those in the NCC were also thought to have formed in situ.

For the ~1.63 Ga mafic dykes, rock magnetic and paleomagnetic experiments attest to magnetite or titanomagnetite as a main magnetic carrier which is also in consistence with petrology of diabase in which magnetite is often the main stable magnetic carrier. Primary remanent magnetization carried by magnetite is believed to be more reliable, compared to that carried by hematite especially in basalt or diabase (Halls et al, 2000; Zhang et al, 2012a,b; Chen et al, 2013; Salminen et al., 2014, 2017; Xu et al, 2014).

Maximum angular deviation (MAD) with a standard of less than 15° is employed as a filter to evaluate the quality of the site-mean directions. Site MLG6 is abandoned because of a large $\alpha_{95}$ value (Table 1). Specimens of sites BTJ and HL from the eastern part of Zoucheng City all exhibit reversed polarity ($D/I$ = 267.5°/−47.9°, $\kappa$ = 80.2, $\alpha_{95}$ = 10.2°, $N$ = 4, including sites BTJ (HL)) and assemble at a declination of ~270° as calculated by PCA or the best-fit great circles method. Thus, it's believable that this reversed polarity has such a direction; however, other nearby sites (sites MLG and XF) present normal polarity with a $D/I$ of 84.8°/54.5° ($\kappa$ = 30.1, $\alpha_{95}$ = 14.2°, $N$ = 5, including sites MLG and XF). The measured normal declination is a slightly higher whilst the inclination is slihtly shallower, compared to dyke DY (Hou et al., 2009). These two directions of oppostie polarity pass the C-classification reversal test of McFadden and McElhinny (1990) at the 95% confidence level ($R$ = 0.101138, critical $R$ at 95% =0.534127, averages Gamma = 6.8 < critical Gamma = 16.5). Because SD or PSD (titano)magnetite in our samples is predominant rather than MD grains, this eliminates magnetic grain size as a cause if the discrepancy observed between the two compasses. In contrast, we observe the intensity of NRM of the 160 samples measured ranges from $3.83 \times 10^2$ A/m to $2.10 \times 10^{-3}$ A/m with only 5 samples whose NRM intensity is $< 1 \times 10^{-1}$ A/m. Thus since most samples from both the ~1.68 Ga and ~1.63 Ga dykes are $> 1 \times 10^{-1}$ A/m (several orders of magnitude bigger than sedimentary rocks), the large difference between the compasses can attribute to the strong NRM intensity of these samples rather than the grain size of (titano)magnetite.

In summary, we interpret these two directions as primary remanent magnetization acquired by dykes during cooling. Finally, a mean direction ($D/I$) for the ~1.63 Ga dykes is 86.1°/53.5° ($\kappa$ = 43.3, $\alpha_{95}$ = 7.9°, $N$ = 9) (Fig. 5a) due to the principle of proximity.

We first calculate the virtual geomagnetic pole (VGP) based on the site-mean directions obtained from 9 sites and then average them to determine the paleomagnetic pole. These site-mean directions from the ~1.63 Ga dykes were taken into account except for MLG6. The corresponding paleomagnetic pole is calculated at 20.8°N, 182.5°E ($\kappa$ = 28.3 $A_{95}$ = 8.3°, $N$ = 9) (Table 1). This paleomagnetic pole also passes the examination of secular variation of geomagnetic following Deenen et al. (2011, 2014) (the value of $A_{95}$ of ~1.63 Ga paleomagnetic pole lies between $A_{95max}$ and $A_{95min}$ which are calculated following Deenen et al., 2011, 2014). From this analysis, a paleo-latitude of 34.05°N was obtained for NCC at ~1.63 Ga.

To summarize, the paleomagnetic pole from the ~1.63 Ga mafic dykes in the Western Luxi area fulfills Van der Voo value Q = 6 (the first six criteria) (Van der Voo, 1990). This newly obtained high-quality Mesoproterozoic paleomagnetic pole at ~1.63 Ga fills in a critical gap

**Table 2**
Available paleomagnetic poles from North China Craton, Baltica, North Australian Craon, Laurentia, Siberia and India.

| Code | Rock unit | Age (Ma) | $P_{lat}$ (°N) | $P_{long}$ (°E) | $A_{95}$ (Dp/Dm) (°) | Test | Reference |
|---|---|---|---|---|---|---|---|
| *North China Craton* | | | | | | | |
| NC1 | Xiong'er Group | ~1780 | 50.2 | 263.0 | 4.5 | R | Zhang et al., 2012a |
| NC2 | Taihang dykes; Yinshan dykes | ~1770 | 36.0 | 247.0 | 2.0 | B | Halls et al., 2000; Xu et al., 2014 |
| NC3 | Western Luxi dyke swarms | ~1630 | 20.8 | 182.5 | 8.3 | R | This study |
| NC4 | Yangzhuang Formation | ~1500 | 2.4 | 190.4 | | F, R | Pei et al, 2005 |
| NC5 | Yangzhuang Formation | ~1500 | 17.3 | 214.5 | 8.0/4.1 | F, R | Wu et al, 2005 |
| NC6 | Tieling Formation | ~1437 | 11.6 | 187.1 | 8.1/4.9 | F | Wu, 2005 |
| NC7 | Yao-liao (Liaoning) sills | ~1350 | −5.9 | 179.6 | 4.3 | F | Chen et al., 2013 |
| *Baltica* | | | | | | | |
| B1 | Keuruu dolerite dykes | 1870 ± 9; 1867 ± 8 | 45.7 | 230.9 | 5.5 | B, R* | Klein et al., 2016 |
| B2 | Småland intrusions | ~1780 | 45.7 | 182.8 | 8.0 | B | Pisarevsky and Bylund, 2010 |
| B3 | Hoting grabbro | ~1780 | 43.0 | 233.3 | 10.9 | B | Elming et al., 2009 |
| B4 | Shoksha Formation | ~1780–1760 | 39.7 | 221.1 | 5.5/2.9 | R | Pisarevsky and Sokolov, 2001 |
| B5 | Turinge grabbro | ~1700 | 51.6 | 220.2 | 4.8 | B | Elming et al., 2019 |
| B6 | Häme dykes | 1642 ± 2; 1647 ± 2 | 23.6 | 209.8 | 14.7 | B, R* | Salminen et al., 2017 |
| B7 | Subjotnian quartz porphyry dykes | ~1630 | 29.0 | 177.0 | 6.0 | | Buchan et al., 2000 |
| B8 | Satakunta sandstone | ~1600 | 27.8 | 173.2 | 6.5 | R, T, B | Klein et al., 2014 |
| B9 | Åland dykes | 1575.9 ± 3 | 23.7 | 191.4 | 2.8 | R, B | Salminen et al., 2016 |
| B10 | Bunkris-Glysjön-Öje dykes | 1469 ± 9 | 28.3 | 179.8 | 13.2 | | Pisarevsky et al, 2014 |
| B11 | Lake Ladoga mafic rocks | 1452 ± 12 | 15.2 | 177.1 | 5.5 | B, R | Lubnina et al., 2010 |
| B12 | Mashak suite | ~1386 | 1.8 | 193.0 | 14.8 | B, R | Lubnina, 2009 |
| B13 | Post Jotnian intrusions | ~1265 | 4.0 | 158.0 | 4.0 | B | Pesonen et al., 2003 |
| *Laurentia* | | | | | | | |
| L1 | Molson dykes (B component) | ~1880 | 27.0 | 219.0 | 4.0 | B | Halls and Heaman, 2000 |
| L2 | Cleaver dykes | ~1740 | 19.4 | 276.7 | 6.1 | B | Irving et al., 2004 |
| L3 | Western Channel dolerite dykes | ~1590 | 9.0 | 245.0 | 6.0 | B | Irving et al., 1972, Hamilton and Buchan, 2010 |
| L4 | St. Francois Mnts. | 1476 ± 16 | −13.2 | 219.0 | 1.7 | G, B, F | Meert and Stuckey, 2002 |
| L5 | Michikamau intrusion | 1460 ± 5 | −1.5 | 217.5 | 4.7 | B, R | Emslie et al., 1976 |
| L6 | Tobacco Root Mnts. Dykes | 1448 ± 49 | 8.7 | 216.1 | 10.3 | B | Harlan et al., 2008 |
| L7 | Zig-Zag Dal intrusions | 1382 ± 2 | 11.0 | 229.0 | 3.0 | R | Evans and Mitchell, 2011 |
| L8 | Mackenzie dykes | 1267 ± 2 | 4.0 | 190.0 | 5.0 | B | Buchan and Halls, 1990 |
| *Siberia* | | | | | | | |
| S1 | Lower Akitkan | 1878 ± 4 | 30.8 | 278.7 | 3.5 | B | Didenko et al., 2009 |
| S2 | Upper Akitkan | 1863 ± 9 | 22.1 | 277.5 | 5.2 | B | Didenko et al., 2009 |
| S3 | Fomich river dykes | 1513 ± 51 | −19.0 | 78.0 | 6.0 | | Veselovskiy et al., 2006 |
| S4 | West Anabar intrusions | 1503 ± 2 | 25.3 | 241.4 | 4.6 | B | Evans et al., 2016 |
| S5 | North Anabar intrusions | 1483 ± 17 | 23.9 | 255.3 | 7.5 | | Evans et al., 2016 |
| S6 | Olenëk mafic intrusions | 1473 ± 24 | 33.6 | 253.1 | 10.4 | B | Wingate et al., 2009 |
| S7 | Chieress dyke | 1384 ± 2 | 5.0 | 258.0 | 6.7 | | Ernst and Buchan, 2000 |
| *North Australia Craton* | | | | | | | |
| NAu1 | Elgee-Pentecost combined | ~1800 | −5.4 | 211.8 | 2.0 | F | Schmidt and Williams, 2008; Li, 2000 |
| NAu2 | Peters Creek volcanics | ~1725 | −26.0 | 221.0 | 4.8 | G, R | Idnurm, 2000 |
| NAu3 | Fiery Creek Fm. | 1709 ± 3 | −23.9 | 211.8 | 10.4 | | Idnurm, 2000 |
| NAu4 | West Branch volcanics | 1709 ± 3 | −15.9 | 200.5 | 11.3 | G | Idnurm, 2000 |
| NAu5 | Tooganinie Fm. | 1648 ± 3 | −61.0 | 186.7 | 6.1 | F, R | Idnurm and Gidings, 1995 |
| NAu6 | Emmerugga dolomite | 1645 | −79.1 | 202.6 | 6.1 | F | Idnurm and Gidings, 1995 |
| NAu7 | Balbirini dolomite (lower) | 1613 ± 4 | −66.1 | 177.5 | 5.7 | R* | Idnurm, 2000 |
| NAu8 | Balbirini dolomite (upper) | 1589 ± 3 | −52.0 | 176.1 | 7.5 | R* | Idnurm, 2000 |
| *India* | | | | | | | |
| I1 | Bastar dykes and Cuddapah sills | ~1880 | 31.0 | 330.0 | 1.9 | R | Meert et al., 2011 |
| I2 | Gwalior traps | ~1789 | 15.4 | 173.2 | 2.0 | | Pradhan et al., 2010 |
| I3 | Newer dolerite dykes | 1765 ± 1 | 45.0 | 311.0 | 1.9 | B | Shankar et al., 2018 |
| I4 | Tiruvannamalai | 1650 ± 10 | −19.0 | 235.0 | 9.0 | | Radhakrishna and Joseph, 1996 |
| I5 | Lakhna dykes | 1466.4 ± 2.6 | 36.6 | 132.8 | 1.3 | R | Pisarevsky et al., 2013 |

Notes: $P_{lat}$-latitude of paleomagnetic poles; $P_{long}$-longitude of paleomagnetic poles; $A_{95}$- 95% circle confidence of paleomagnetic poles; R-reversal test; B-baked test; F-fold test; G-conglomerate test; R*-dual polar directions; T- tilt test.

for the apparent polar wander pole (APWP) of NCC during the Paleo-MesoProterozoic, which is essential for Precambrian continental reconstruction. Therefore, we suggest this newly obtained pole as a 'key' pole (Buchan, 2013).

### 6.2. Paleomagnetic pole of North China Craton at ~1.68 Ga

The ~1.68 Ga dykes in the Eastern Luxi area, sites ZJL, XSD and YD display normal polarity. Sites ZJL and XSD from the eastern part of Laiwu City, they display normal polarity and even the directions of sites ZJL4 and XSD2 overlap within the 95% confidence circle. Futhermore, site YD from the northern part of Linyi City also has a normal polarity and its direction is consist with sites ZJL and XSD (Fig. 5b). An overall site-mean direction (*D/I*) for the ~1.68 Ga dykes is 89.1°/47.1° ($\kappa$ = 35.0, $\alpha_{95}$ = 13.1°, $N$ = 5) (Fig. 5b), which is apparently inconsistent with the ~1.63 Ga dykes though the difference within 7° still exist, especially in terms of inclination. No isolated reversed directions are observed; however, this normal direction seems to be antipodal to the reversed one at ~1.63 Ga.

The corresponding pole is calculated at 17.8°N, 184.9°E ($\kappa$ = 29.6 $A_{95}$ = 14.3°, $N$ = 5) and this pole also passes the examination of secular variation of geomagnetic field following Deenen (2011, 2014). Specifically, the paleo-latitude of NCC at ~1.68 Ga was 28.28°N. An angle deviation of 5.77° between ~1.63 Ga and ~1.68 Ga is evident, but the

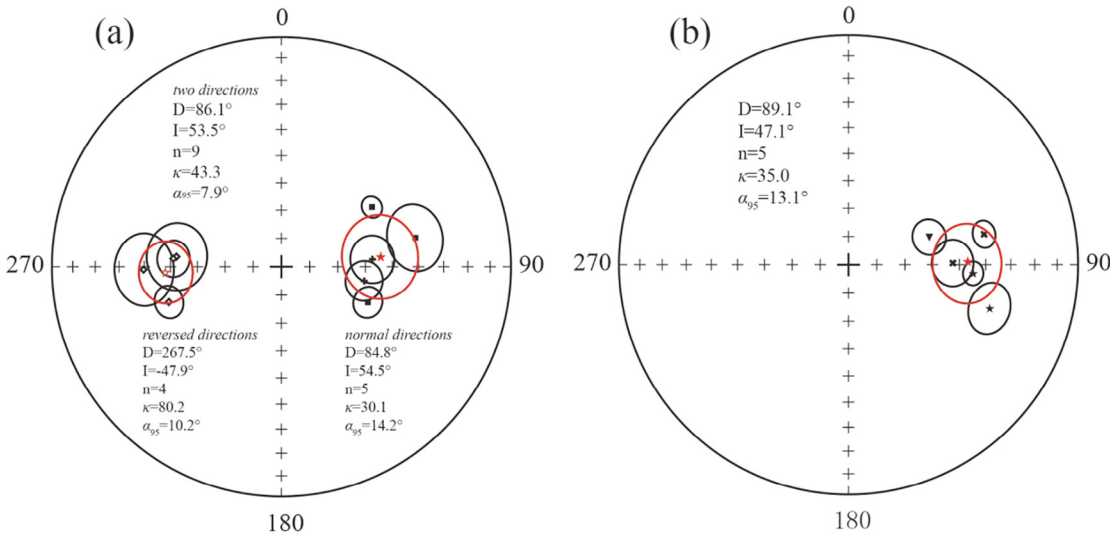



**Fig. 5.** Equal-area projections of both normal and reversed directions isolated from mafic dykes in the Luxi area. (a) Both normal and reversed directions isolated from ~1.63 Ga mafic dykes in the Western Luxi area. (b) Normal directions isolated from ~1.68 Ga mafic dykes in the Eastern Luxi area. Open (closed) symbols represent reversed (normal) inclination. Diamond, square, cross, oblique cross, black 5-pointed star and triangle down symbol represent sample sites BTJ (HL), MLG, XF, XSD, ZJL and YD, respectively. Closed red 5-pointed star represents positive site-mean directions isolated from ~1.63 Ga and ~1.68 Ga mafic dykes, whereas the open red 5-pointed star means the negative site-mean direction calculated from ~1.63 Ga mafic dykes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

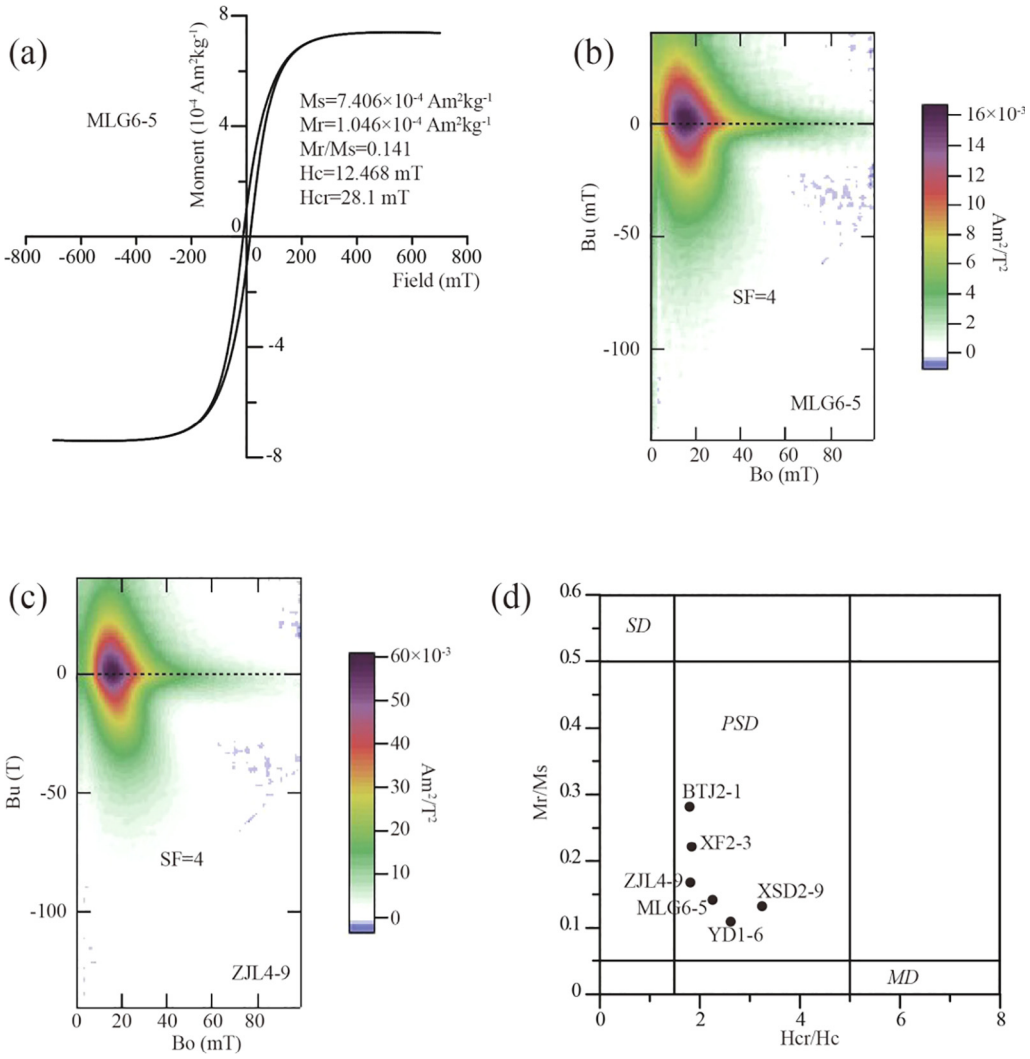



**Fig. 6.** (a) Narrow-waisted hysteresis loop for a representative sample MLG6-5 from the ~1.63 Ga mafic dykes in the Western Luxi area. (b-c) First order reversal curves (FORCs) show the single domain structure of magnetite in samples MLG6-5 (~1.63 Ga) and ZJL4-9 (~1.68 Ga). (d) Five representative samples are plotted in the Day diagram (Day et al., 1977; Dunlop, 2002) to demonstrate the domain state of (titano) magnetite. Ms, Mr, Hc and Hcr represent saturation magnetization, saturation remanent magnetization, coercive force and coercivity of remanence, respectively. SD, PSD, MD in the Day diagram stand for single domain, pseudo-single domain and multi-domain, respectively.

key implication is that the NCC was positioned at the mid-low latitudes from ~1.68 Ga to ~1.63 Ga.

Since the site-mean directions exhibit minimal dispersion (Fig. 5; Table 1), it follows that the spatio-temporal formation of the dykes had no substantive deleterious on the quality of paleomagnetic results. In conclusion, there was apparently little movement of the NCC from ~1.68 to 1.63 Ga.

### 6.3. Proterozoic APWP of North China Craton

We obtained two Mesopreterozoic paleomagnetic pole for North China Craton at ~1.63 Ga and ~1.68 Ga which broadly represent two phases of mafic dykes intrusions. Since the pole at ~1.68 Ga failed to pass both reversal test and field test, we place greater emphasis on the reliability of the ~1.63 Ga paleomagnetic pole to represent the apparent polar wander path (APWP).

Reliable paleomagnetic poles with accurate age control are used to determine the apparent polar wander path (APWP), by which the tendency of continental movement is estimated. There are several available high-quality paleomagnetic poles from the NCC (Table 2) to establish the APWP (Zhang et al., 2012a,b; Halls et al., 2000; Pei et al., 2005; Wu et al., 2005; Wu, 2005; Chen et al., 2013). The first paleomagnetic studied by Halls et al. (2000) yielded a key pole at ~1.77 Ga. Later, both Zhang et al. (2012) and Xu et al. (2014) have reported other two paleomagnetic poles of North China Craton, providing useful spatial constrains. The age of Yangzhuan Formation was originally proposed to be ~1.35 Ga but has recently been revised to be ~1.5 Ga (Zhang et al., 2009; Li et al., 2010; Pei et al., 2005; Wu et al., 2005). Wu (2005) usefully obtained an available pole at ~1.44 Ga from the Tieling Formation, whilst Chen et al. (2013) obtained an additional pole at ~1.35 Ga which also passed a fold test and was therefore used in our study.

We drew a more detailed APWP by integrating our new results with previous data and aim to provide a refined APWP for NCC during Paleo-Mesoproterozoic. This new pole at ~1.63 Ga is plotted along the expected path as delimited by previous poles. Previous studies (e.g. Xu et al., 2014) proposed the view that Laurentia had a closely related with NCC, which may result from a lack of paleomagnetic poles during the past 0.3 Ga; however, the paleomagnetic pole at ~1.63 Ga effectively distinguishes NCC compared to Laurentia to elucidate the spatio-temporal characteristics of the cratons adjacent to NCC as discussed in the next section.

### 6.4. Implications for the reconstruction of North China Craton within the Columbia supercontinent

Several models have located the NCC in different positions within *the Columbia* (Chen et al., 2013; Halls et al., 2000; Hou et al., 2008; Peng et al., 2008; Pei et al., 2005; Wu et al., 2005; Zhang et al., 2009; Zhang et al., 2012a; Zhao et al., 2002, 2005), but our new results from the Luxi area permit an APWP-constrained reconstruction of the location of the NCC within this supercontinent. Reliable Paleo-Mesoproterozoic paleomagnetic poles (which could pass field test or reversal test and had a lower value of $A_{95}$, less than 15°) from ~1.8 to 1.3 Ga were taken into account to achieve this aim. Sketches of the APWP of potential cratons which had geological affinity with the NCC are shown in Figs. 7 and 8.

The similarity of the APWPs unambiguously testify to the connection between Baltica, Laurentia and Siberia (Evans and Mitchell, 2011). In terms of the NCC, its shape presents as an curved line from codes NC1 (~1.78 Ga) to NC7 (~1.35 Ga), comparable to that of Baltica, Laurentia and Siberia. Moreover, it's clear that the NCC has drifted from high- to lower latitude (Fig. 7a) during the Paleo-Mesoproterozoic, consistent with the other major cratons in *the Columbia*. These confirm NCC has been a part of *the Columbia* during Paleo-Mesoproterozoic; however, the two inflexions occur at codes NC3 (~1.63 Ga) and NC4 (~1.5 Ga),

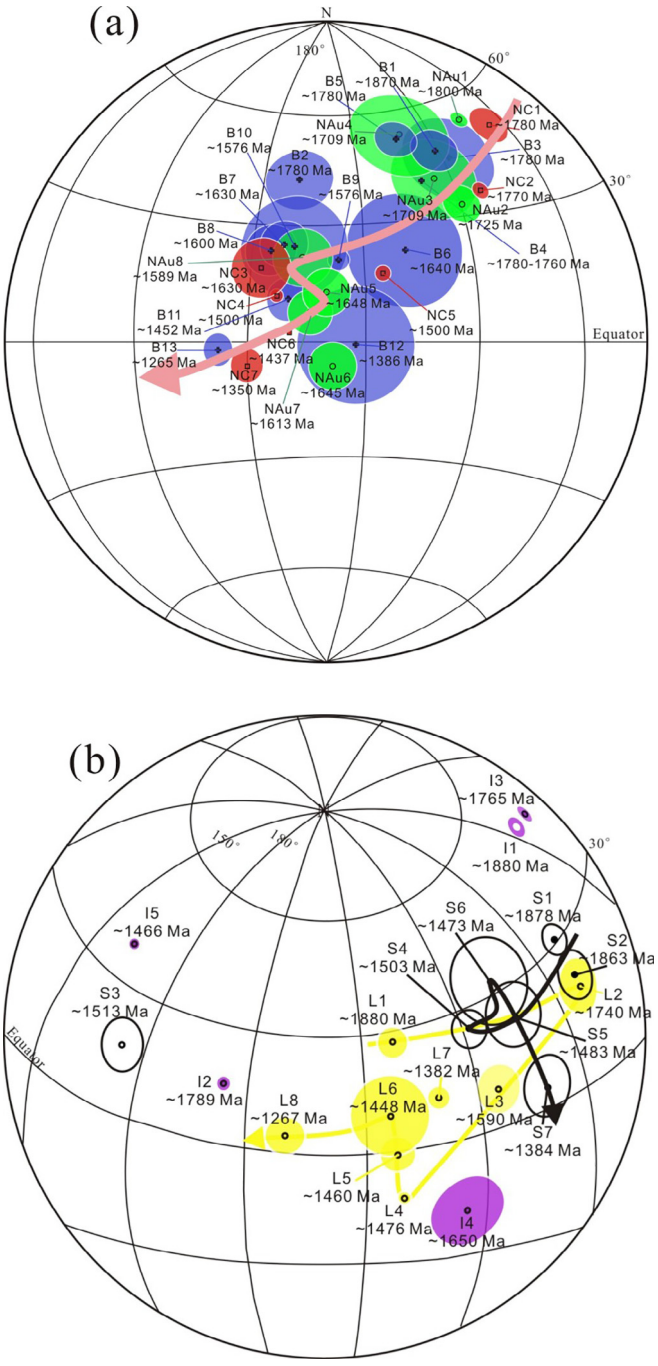



**Fig. 7.** Equal-area projections to display available Paleo-Mesoproterozoic paleomagnetic poles for (a) Baltica (blue), North Australia Craton (green), North China Craton (red); (b) Laurentia (yellow), Siberia (black) and India (purple) and corresponding apparent polar wander path (APWP) for each of the continents. In Fig. 7a, the paleomagnetic poles of Baltica and the North Australia Craton were rotated to the frame of the North China Craton to examine the possible paleomagnetic affinity of these three cratons. Euler rotation parameters are: Baltica, 10°, 100°, 75°; North Australia Craton, 63°, 220°, 16° but there is no Euler rotation of paleomagnetic poles for Laurentia, Siberia and India, as shown in Fig. 7b. The pink line in Fig. 7a represents the potential tendency of the movement pattern of Baltica, North China Craton and North Australia Craton during the Paleo-Mesoproterozoic. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

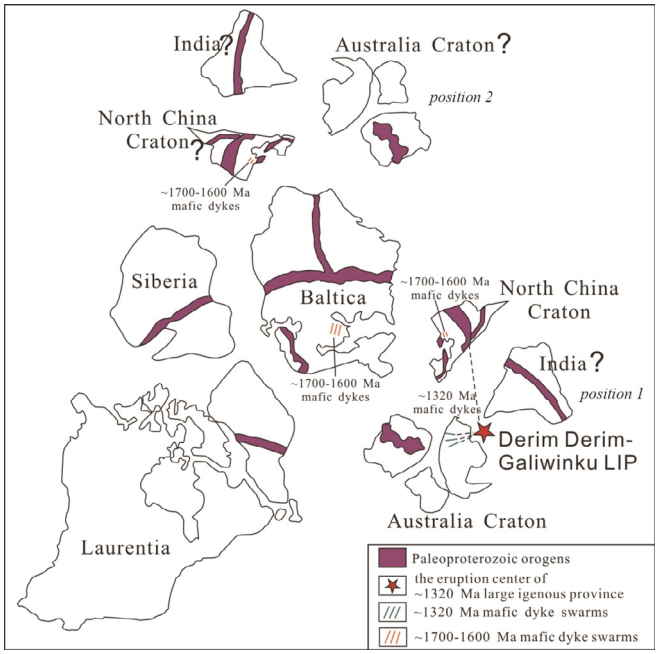



**Fig. 8.** Reconstruction map of the North China Craton and potential adjacent cratons (Baltica and North Australia Craton). All reconstructions are shown in the present geographic coordinate system. Laurentia is fixed to its contemporary geographic location and other cratons are rotated to the frame of Laurentia. Euler rotation parameters are: Siberia, 75.95°,111.67°, 153.71°; Baltica, 53.51°, 4.83°, 45.60°; North China Craton, −13.03°, −129.25°,-98.64° (*position 1*) or −48.09°, −78.01°, −142.27° (*position 2*); North Australia Craton, −63.96°, −13.86°, 181.54° (*position 1*) or −46.44°, −2.62°, 94.56° (*position 2*); India, 43.82°, 34.79°, −152.46° (*position 1*) or 37.89°, 85.01°, 128.18° (*position 2*).

forming the lightening-like shape after code NC3 (~1.63 Ga) (Fig. 7a), which distinguishes it from Laurentia and Siberia. Concerning the fact that paleo-longitude cannot be calculated using paleomagnetic data, if we rotate the paleomagnetic poles of Baltica to the frame of the NCC to examine their similarity, the APWPs of the NCC and Baltica are almost completely identical, providing the evidence for a paleomagnetic connection between them (Fig. 7a). Coeval dyke swarms have been reported in Finland (Salminen et al., 2014, 2017, 2018), which may clarify how the northern margin of the NCC was linked to the southeastern edge of Baltica (Fig. 7). In addition, later mafic intrusions (~1.4 to 1.3 Ga) also occurred in Siberia, Laurentia and Australia (Ernst et al., 2008). The similarities between orogenic events (Wilde et al., 2002) and the overall geological evolution support the hypothesis that the NCC was connected with Baltica. In addition, binned frequency histograms of detrital zircon ages attest to the possible linkage between the northern NCC and Baltica (Liu et al., 2014). A similar pattern of southwest movement is evident for Laurentia and Siberia up until ~1.5 Ga when they begun to drift northward. This significantly differs from the pattern of movement exhibited by the NCC, North Australian craton and Baltica which moved southward at ~1.5 Ga.

To conclude, the location of NCC within *the Columbia* was comfirmed by the APWP. The similarity between the APWP of the NCC and Baltica indicates these two continents were connected with each other during the Paleo-Mesoproterozoic (Wilde et al., 2002; Pei et al., 2005).

Zhang et al. (2017, 2018) proposed a scenario in which the northern NCC and North Australia Craton were linked together based on stratigraphically coeval black shales identified in both cratons. Although only a few paleomagnetic poles were reported in late Mesoproterozoic, the APWP of the North Australia Craton reconstructed from the early Mesoproterozoic paleomagnetic poles exhibits almost a very similar trends to the NCC. By extrapolating this movement trend using codes NAu1 (~1.8 Ga) to NAu5 (~1.648 Ga), we discovered that the North Australia Craton had moved southward, similar to the other continents. At code NAu6 (~1.65 Ga), however, a turning point occured whereby the North Australia Craton suddenly moved toward southeast. A second turning point at NAu7 (~1.613 Ga) then indicates a potential westerly direction. This APWP was very similar to that of the NCC with reference to the lightening-like shape with contemporaneous turning points at NC3 (~1.6 Ga) and NAu6 (~1.65 Ga). Therefore, we propose that there was a spatio-temporal connection between the NCC and the North Australia Craton during the Mesoproterozoic.

Zhao et al. (2002, 2005) initially demonstrated the linkage between the northern NCC and eastern India based on metamorphic petrology. Although the location of the NCC and India within *the Columbia* differs from that of Zhao et al. (2002, 2005), the connection between them is supported by nearly simultaneous magmatism in both continents, for instance, Xiong'er volcanic rocks in NCC (Peng et al., 2008) and coeval dyke swarms in India and the NCC (Hou et al., 2008). An APWP was not constructed for India due to dearth of paleomagnetic data. Notably, however, the paleomagnetic poles attest to a southward- followed by northward trajectory similar to the other continents (Fig. 7b). Thus, we conclude that the India may have been a part of *the Columbia* but more paleomagnetic data is needed to permit an accurate comparison with the other continents.

## 7. Conclusions

New high-quality ~1.63 Ga paleomagnetic pole utilizing 90 specimens from mafic dyke swarms in Western Luxi area yielded a mean direction of 20.8°N, 182.5°E ($A_{95}$ = 8.3°, $N$ = 9) for the North China Craton (NCC). This result fulfills a Van der Voo value Q = 6 (the first six criteria) (Van der Voo, 1990). The ~1.63 Ga paleomagnetic pole constrained the location of NCC at 34.05°N. In addition, we also obtained a paleomagnetic pole at ~1.68 Ga from mafic dykes in Eastern Luxi area, with a mean pole direction at 17.8°N, 184.9°E ($A_{95}$ = 14.3°, $N$ = 5) for the NCC. This pole also passes the test of secular variation of geomagnetic field. The addition of the new ~1.63 Ga pole integrated with other reliable Mesoproterozoic paleomagnetic poles of NCC permit a signigicant improved APWP. This updated APWP allows a more robust comparison to be drawn between the NCC and other major Precambrian cratons. The fact that the NCC APWP matches well with these other cratons (Baltica, Laurentia and Siberia) is a clear indication that NCC was part of *the Columbia* at 1.6 Ga; however, evidence for a turning point at ~1.60 Ga was highlighted post-1.6 Ga whereby the NCC, Baltica and North Auatralia Craton migrated south, but Laurentia and Siberia moved north at ~1.5 Ga. In addition, the APWP of North Australia Craton shared almost the same kinematic tendency with NCC, together with geological evidences like black shales (Zhang et al., 2018) to support a connection between northern NCC and North Australia Craton during the Mesoproterozoic. The mechanisms responsible for these different directions of movement are, at present, elusive. Thus, more investigations are required to probe crucial geodynamic evolution in this time interval. In addition, the proposed relation between the southern NCC and eastern India is tenuous and requires further investigations.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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## Appendix A. Supplementary data

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